WILL Part I (addition)

Closed Algebraic System of Relational Orbital Mechanics (R.O.M.)

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1 Closed Algebraical System of R.O.M. Equations

Dedicated sections in WILL RG Part I: https://willrg.com/documents/WILL RG I.pdf-sec:rom

Desmos Project

R.O.M.

Remark 1.1. *R.O.M. does not describe how a body moves under forces; it classifies the algebraically allowed relational states of a bound system. It is not equations of motion but algebraically closed system of allowed states within WILL - allowed free will.*

$$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}} \; (z_\kappa = \text{gravitational redshift})$$

$$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}} \; (z_\beta = \text{transverse Doppler shift})$$

Observational Inputs

$Z_{sys} = (1 + z_\kappa)(1 + z_\beta) = \frac{1}{\tau}$ (product of gravitational red shift and transverse Doppler shift)
$\tau = \kappa_X \cdot \beta_Y = \frac{1}{Z_{sys}}$ (product of projectinal phase factors on S^1 and S^2 carriers)
$z_\kappa = \frac{1}{\kappa_X} - 1$ (gravitational redshift)
$z_\beta = \frac{1}{\beta_Y} - 1$ (transverse Doppler shift)
$z_{\kappa o} = \frac{1}{\kappa_{Xo}} - 1$ (redshift at phase o)
$z_{\beta o} = \frac{1}{\beta_{Yo}} - 1$ (transverse Doppler shift at phase o)

*This work is archived on Zenodo: ¸

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Global System Parameters (Fixed for the Orbit)

$\kappa = \sin(\theta_2) = \sqrt{1-(1+z_\kappa)^{-2}} = \sqrt{\frac{R_s}{a}} = \sqrt{\frac{\rho}{\rho_{max}}} = \sqrt{2}\left(\frac{\pi R_s}{Tc}\right)^{\frac{1}{3}} = \sqrt{\kappa_p^2(1-e)} = \sqrt{4W} = \sqrt{2(\kappa_o(o)^2-\beta_o(o)^2)} == \sqrt{\frac{1}{2}(3-\sqrt{1+8\tau_{Wo}(O_o)^2})} = \kappa_R^2(\frac{I_t}{T})^{\frac{2}{3}} = \sqrt{\kappa_o^2\cdot\eta_o}$ (potential projection at semi-major axis)
$\kappa_X = \cos(\theta_2) = \sqrt{1-\kappa^2} = \frac{1}{(1+z_\kappa)}$ (potential phase factor)
$\kappa_R = \sqrt{\frac{\kappa^2}{\sin(\theta_R)}} = \sqrt{\frac{R_s}{\frac{T}{2\pi}c\beta\sin(\theta_R)}}$ (potential projection at the surface of)
$\beta = \cos(\theta_1) = \sqrt{1-(1+z_\beta)^{-2}} = \frac{\kappa}{\sqrt{2}} = \beta_p\sqrt{\frac{1-e}{1+e}} = \sqrt{2W} = \frac{2\pi a}{Tc} = (\frac{\pi R_s}{Tc})^{\frac{1}{3}} = \sqrt{(\kappa_o(o)^2-\beta_o(o)^2)} = \sqrt{\frac{\kappa_p^2}{2}(1-e)} = \sqrt{\kappa_o^2-\frac{\kappa_o^2}{2}\cdot\left(1+\left(\frac{1}{\delta_o(o)}-1\right)\right)} = \beta_o(o)\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\cdot e\cdot\cos(o)}}$ (kinetic projection on semi major axis)
$\beta_Y = \sin(\theta_1) = \sqrt{1-\beta^2}$ (kinetic phase factor)
$\beta_{int} = \frac{\beta}{\sqrt{1-e^2}}$ (interior kinetic projection)
$\tau = \kappa_X\beta_Y$ (relational spacetime factor)
$\tau_Y = \sqrt{1-\tau^2} = \sqrt{3\beta^2-2\beta^4}$ (relational spacetime phase factor)
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{\frac{3}{2}}\ \kappa = \sqrt{3}\ \beta$ (total relational shift (magnitude of state difference))
$R_s = \kappa_o(o)^2 r_o(o) = \frac{2Gm_0}{c^2} = \frac{\Delta_{to}(o)}{\zeta(o)}\kappa^2\beta c = Tc\frac{\beta^3}{\pi} = t_o(o)\kappa_o(o)^2\cdot c = \frac{r_1 r_2}{r_2-r_1}(\beta_1^2-\beta_2^2) = \frac{a}{2}(3-\sqrt{1+8\tau_{Wo}(O_o)^2}) =$
$= \frac{r_o(o)}{2(2a-r_o(o))}(4a-r_o(o)-\sqrt{(4a-r_o(o))^2-8a(2a-r_o(o))(1-\tau_{Wo}(o)^2)}) = \frac{T}{\pi}(\sqrt{1-(\frac{1}{1+z_\beta})^2})^3 c$
(Schwarzschild radius - system scale)
$a = \frac{R_s}{\kappa^2} = \frac{R_s}{4W} = \frac{\kappa cT}{2\pi\sqrt{2}} = \frac{\beta_o c}{\omega}\cdot\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e\ \cos(o)}} = \frac{\Delta_{to}(o)}{\zeta}\beta c$ (semi-major axis)
$R_{surf} = a\cdot\theta_R$ (physical radius)
$m_0 = \frac{\kappa^2c^2a}{2G} = 4\pi\rho a^3$ (mass parameter)
$t = \frac{a}{c}$(temporal scale of the system)
$W = \frac{\beta^2}{2} = \frac{1}{2}(\kappa_o^2-\beta_o^2) = \frac{1}{4}\kappa_p^2(1-e) = \frac{1}{2}(\kappa_o(o)^2-\frac{\kappa_o(o)^2}{2\delta_o(o)}) = \frac{1}{2}((1-(1+z_{\kappa o}(o))^{-2})-(1-(1+z_{\beta o}(o))^{-2})) = (\frac{\pi R_s}{2\sqrt{2}cT_o})^{\frac{2}{3}}$ (energy invariant - binding energy)
$\Delta\phi = \frac{2\pi\cdot\tau_Y^2}{1-e_\alpha^2} = \frac{2\pi(3\beta^2-2\beta^4)}{1-e^2}$ (precession of perihelion per orbit)
$h_W = r_o(o)\beta_T(o)c = r_o^2\cdot\omega_o = a\cdot\beta c\cdot e_Y = R_s c\frac{\sqrt{1-e^2}}{2\beta}$ (invariant angular momentum)
$\omega = \frac{\beta c}{a} = \frac{c}{R_s}2\beta^3$ (angular frequency)
$T = \frac{2\pi}{\omega} = \frac{2\pi a}{\beta c} = \frac{\pi R_s}{c\beta^3}$ (orbital period)
$\Delta\varphi = 2\arcsin(\frac{\kappa_p^2(1+\beta_p^2)}{2\beta_p^2-\kappa_p^2(1+\beta_p^2)})$ (gravitational deflection)
$\Delta\gamma = 2\arcsin(\frac{\kappa_p^2}{\kappa_{Xp}^2})$ (light deflection)

Eccentricity Relations

$e = \frac{1}{\delta_p}-1 = 1-\frac{2\beta_a^2}{\kappa_a^2} = \frac{2\beta_p^2}{\kappa_p^2}-1$ (eccentricity derived from closure)
$e_Y = \sqrt{1-e^2}$ (eccentricity's orthogonal value)
$e_X = \frac{1+e}{1-e} = \frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (shape factor, still looks like magic to me)

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Time-phase

$\omega_o(o) = \frac{\beta \cdot c}{a} \cdot \frac{(1+e \cdot cos(o))^2}{(1-e^2)^{\frac{3}{2}}}$ (angular frequency at phase o)
$\Delta_{to}(o) = \int_0^o \frac{1}{\omega_\theta(\theta)} \, d\theta = \frac{a}{\beta c}\zeta = \frac{T_O}{2\pi}\zeta$ (time duration of given phase interval)
$\zeta = (1-e^2)^{\frac{3}{2}} \cdot (\int_0^o (1+e \cdot \cos(\theta))^{-2} d\theta) = \frac{1}{\sqrt{1-e^2}} \int_0^o (\frac{t_o(\theta)}{t})^2 d\theta$ (Temporal Phase interval)

Perihelion Relations

$r_p = a(1-e) = \frac{R_s}{\kappa_p^2}$ (radius at perihelion)
$\kappa_p = \kappa\sqrt{\frac{1}{1-e}} = \sqrt{\frac{2\beta^2}{1-e}} = Q_p\sqrt{\frac{2}{3+e}}$ (potential projection at perihelion)
$\kappa_{Xp} = \sqrt{1-\kappa_p^2}$ (potential phase projection at perihelion) $\beta_p = \frac{V_p}{c} = \beta \sqrt{\frac{1+e}{1-e}} = \frac{r_p \omega_o(0)}{c} = \frac{\kappa_p}{\sqrt{2}}\sqrt{1+e}$ (kinetic projection at perihelion)
$\delta_p = \frac{\kappa_p^2}{2\beta_p^2} = \frac{1}{1+e}$ (closure factor at perihelion)
$Q_p = \sqrt{\kappa_p^2 + \beta_p^2}$ (relational shift at perihelion)

Aphelion Relations

$r_a = a(1+e) = \frac{R_s}{\kappa_a^2}$ (radius at aphelion)
$\beta_a = \sqrt{\beta_p^2 e_X^{-2}} = \beta\sqrt{e_X^{-1}}$ (kinetic projection at aphelion)
$\kappa_a = \sqrt{2W + \beta_a^2}$ (potential projection at aphelion)
$\delta_a = \frac{1}{1-e} = \frac{\kappa_a^2}{2\beta_a^2}$ (closure factor at aphelion)
$Q_a = \sqrt{\kappa_a^2 + \beta_a^2}$ (relational shift at aphelion)

Phase Variables (Depend on o)

$o =$ orbital phase in radians
$r = r_o(o) = a\frac{1-e^2}{1+e\cos o} = \frac{R_s}{\kappa_o^2}$ (radial distance at phase o)
$\kappa_o = \sqrt{\beta_R(o)^2 + \beta_T(o)^2 + \beta^2} = \sqrt{\frac{R_s}{r}} = 1-(1+z_{\kappa o}(o))^{-2} = \kappa_p\sqrt{\frac{1+e\cos(o)}{1+e}} = \sqrt{\beta^2+\beta_o^2} = \sqrt{\kappa_{surface}^2 \cdot \sin(\theta_{obs})}$ (local potential at phase o)
$\kappa_{Xo} = \sqrt{1-\kappa_o^2} = (z_{\kappa o}(o)+1)^{-1}$ (gravitational phase factor at phase o)
$\beta_o(o) = \beta \cdot \frac{\sqrt{1+e^2+2 \cdot e \cdot \cos(o)}}{\sqrt{1-e^2}} = \sqrt{\kappa_o^2 - 2W} = \frac{\kappa_o(o)}{\sqrt{2\delta_o(o)}} == \sqrt{\frac{\kappa_o(o)^2}{2}\frac{1+e^2+2 \cdot e \cdot \cos(o)}{1+e \cdot \cos(o)}} = \sqrt{\frac{R_s}{r_o(o)} - \frac{R_s}{2a}} = \sqrt{1-(1+z_{\beta o}(o))^{-2}}$ (kinetic projection at phase o)
$\beta_R(o) = \beta \, \frac{e\sin(o)}{\sqrt{1-e^2}} = \sqrt{\beta_o(o)^2 - \beta_T(o)^2} = \sqrt{((1-(1+z_{\kappa o}(o))^{-2}) - 2W) - \frac{r_o(o)^2\omega_o(o)^2}{c^2}}$ (radial kinetic projection at phase o)
$\beta_T(o) = \frac{r_o(o)\omega_o(o)}{c} = \beta\frac{1+e\cos(o)}{\sqrt{1-e^2}} = \kappa_o^2\frac{\sqrt{1-e^2}}{2\beta}$ (transverse kinetic projection at phase o)
$\beta_{Yo} = \sqrt{1-\beta_o^2} = (z_{\beta o}(o)+1)^{-1}$ (relativistic phase factor at phase o)
$\delta_o = \frac{1+e\cos o}{1+e^2+2e\cos o} = \frac{\kappa_o^2}{2\beta_o^2}$ (local closure factor at phase o)
$Q_o = \sqrt{\kappa_o^2 + \beta_o^2}$ (local relational shift vector at phase o)

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$\omega_o = a\beta c\frac{e_Y}{r_o^2} = \frac{\beta c}{a}\frac{(1+e\cos o)^2}{(1-e^2)^{3/2}}$ (angular speed at phase o)
$\eta_o = \frac{r}{a} = 2 - \frac{2\beta_o(o)^2}{\kappa_o(o)^2}$ (phase scale amplitude at phase o)
$\tau(o) = \kappa_{Xo}(o) \cdot \beta_{Yo}(o) = Z_{sys}(o)^{-1}$ (phase spacetime factor at phase o)
$t_o = \frac{r_o(o)}{c}$ (light time scale at phase o)
$\Delta_o = \frac{\tau_Y^2 \cdot o}{1-e^2}$ (precession of perihelion at phase o)

Orbital Phase Invariants

Holographic Decryption Invariant

$$Z_{raw}(-\omega_i)\tau(-\omega_i)(1 - e\cos\omega_i) + Z_{raw}(\pi - \omega_i)\tau(\pi - \omega_i)(1 + e\cos\omega_i) = 2 \qquad (1)$$

$\beta^2 = (\kappa_o^2 - \beta_o^2)$
$h_W = r_o^2 \cdot \omega_o$
$\frac{\beta_T(o)}{\kappa_o^2(o)} = \frac{\sqrt{1-e^2}}{2\beta}$
$\beta_T(o)^2\eta_o^2 = \beta^2(1-e^2)$

Relational Geometry (WILL)

$\theta_1 = \arccos(\beta)$ (distribution angle on S^1, non-physical)
$\theta_2 = \arcsin(\kappa)$ (distribution angle on S^2, non-physical)
$\Delta_Q = Q_o^2 - Q^2$ (phase relational shift amplitude at phase o)
$O_o = \arccos(1 - \delta^{-1}) = \arccos(-e) = \arccos(1 - \frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2} - 1)$ (orbital balance point where $\kappa_o^2 = 2\beta_o^2$ is true)

Structural-Dynamical Equivalence

$\frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{1+e}{1-e} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (remarkable equivalence of structural (κ on S^2 carrier) and dynamical (β on S^1 carrier) symmetry suggesting once again that $SPACETIME \equiv ENERGY$)

Observer dependant

$Z_{raw}(o) = (1 + \beta_{int}(\cos(o + \omega_i) + e\cos(\omega_i))\sin(i))\ Z_{sys}(o)$ (raw light shift including the line of site Doppler at phase o)
$\beta_{los}(o) = \frac{\beta}{\sqrt{1-e^2}}(\cos(o + \omega_i) + e\cos(\omega_i))\sin(i)$ (line of site light shift)
$i =$ (orbital inclination in relation to line of site and orbital plane)
$\omega_i =$ (phase turn or argument of periapsis)
$K_i = \frac{\beta}{\sqrt{1-e^2}}\sin(i) = \beta_{int}\sin(i)$ (semi-amplitude invariant)
$Z_{rawmax} = Z_{sys}(-\omega_i)(1 + K_i(1 + e\cos\omega_i))$
$Z_{rawmin} = Z_{sys}(\pi - \omega_i)(1 + K_i(-1 + e\cos\omega_i))$

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$O_{oi} = (O_o + \omega_i)$
$Z_{sys}(-\omega_i) = (1 - \beta^2 \frac{1+e^2+2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2 \frac{1+e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{sys}(\pi - \omega_i) = (1 - \beta^2 \frac{1+e^2-2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2 \frac{1-e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{raw}(-\omega_i)\tau(-\omega_i)(1 - e\cos\omega_i) + Z_{raw}(\pi - \omega_i)\tau(\pi - \omega_i)(1 + e\cos\omega_i) = 2$ (Holographic Decryption Invariant)

Background Free Parametric Equations for the Observed Orbit

$$\begin{bmatrix} x_{sky} \\ y_{sky} \\ z_{depth} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \begin{bmatrix} x_{orb} \\ y_{orb} \\ 0 \end{bmatrix}$$

$$x_{sky} = r(o)\cos(o + \omega_i)$$
$$y_{sky} = r(o)\sin(o + \omega_i)\cos(i)$$
$$z_{depth} = r(o)\sin(o + \omega_i)\sin(i)$$

Un-tilted 2D coordinates within the orbital plane

$x_{orb} = r(o)\cos(o + \omega_i)$
$y_{orb} = r(o)\sin(o + \omega_i)$

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